

pfSense Firewall Deployment

pfSense Firewall was deployed on Hyper-V to act as a central routing, firewall, and network segmentation device for the enterprise infrastructure lab environment.

The firewall is responsible for controlling traffic between departmental networks and providing controlled Internet access through the WAN interface.

1. pfSense Installation

pfSense was installed as a virtual machine on Hyper-V using the installation ISO image.

Installation Procedure

- 1. Open **Hyper-V Manager**.
- 2. Click **New** → **Virtual Machine**.
- 3. Enter the virtual machine name (pfSense Firewall).
- 4. Assign startup memory.
- 5. Connect first network adapter to **WAN-SWITCH**.
- 6. Attach pfSense ISO image as boot media.
- 7. Start the virtual machine.
- 8. Follow pfSense installation wizard.
- 9. Complete installation and reboot the system.

Result

pfSense Firewall was successfully installed on Hyper-V.

2. Interface Configuration

After installation, pfSense network interfaces were assigned to Hyper-V virtual switches for network segmentation.

Interface Mapping

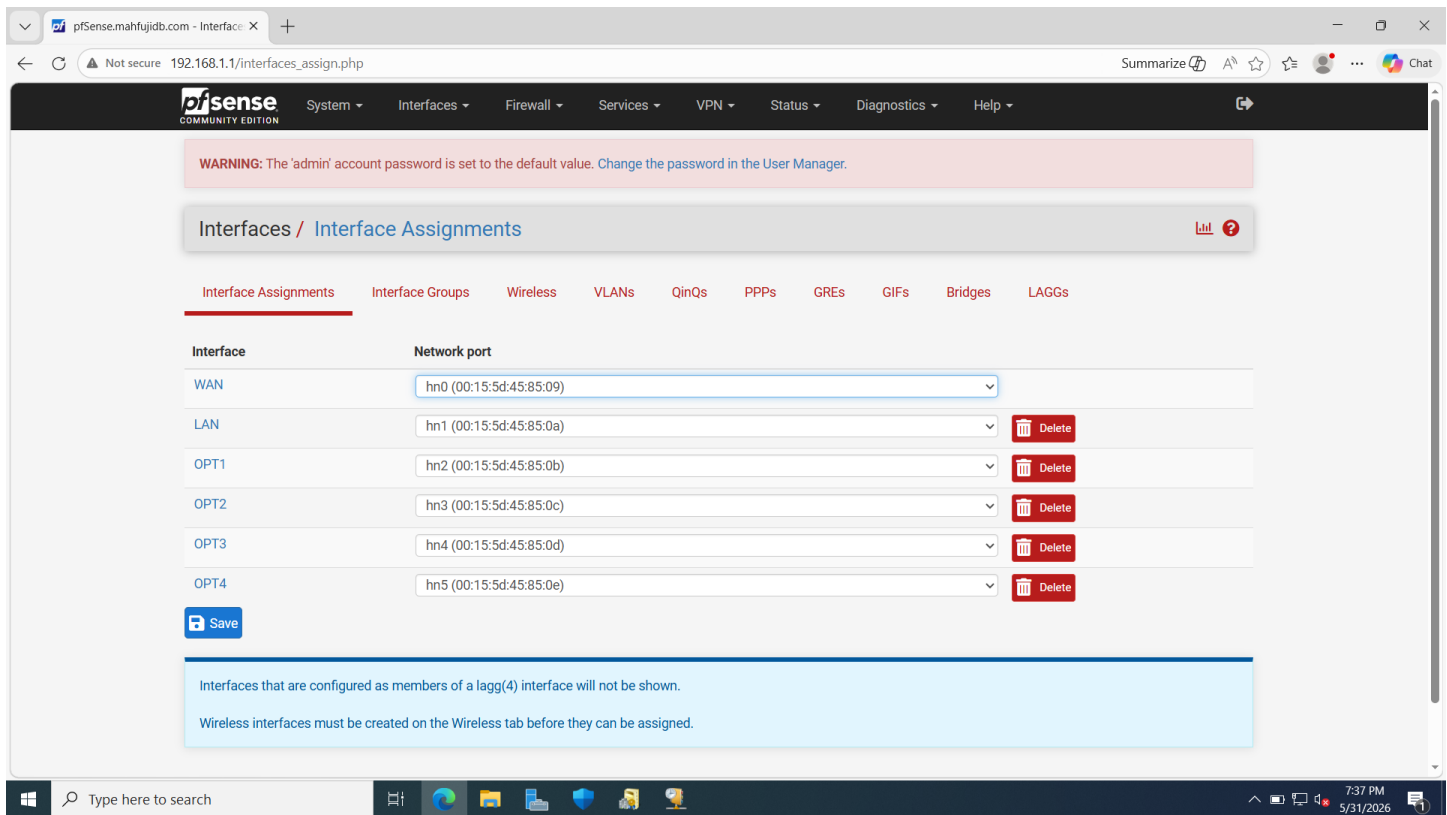
Interface	Hyper-V Switch	IP Address	Mode
WAN (hn0)	WAN-SWITCH	DHCP	Internet
LAN (hn1)	SERVER-SWITCH	192.168.1.1	Internal Gateway
OPT1 (hn2)	MAMT-SWITCH	192.168.2.1	Management Network
OPT2 (hn3)	FIN-SWITCH	192.168.3.1	Finance Network
OPT3 (hn4)	HR-SWITCH	192.168.4.1	HR Network
OPT4 (hn5)	IT-SWITCH	192.168.5.1	IT Network

Interface Assignment Procedure

1. Access pfSense console.
2. Select **Assign Interfaces**.
3. Map network adapters (WAN, LAN, OPT1–OPT4).
4. Confirm assignments.
5. Set IP addresses for each interface.
6. Save configuration.

Result

All pfSense interfaces were successfully assigned and configured with proper IP addressing.



3. Network Segmentation Configuration

pfSense was configured as the central firewall and gateway for segmented departmental networks.

Network Design

Network	Department	Gateway
192.168.1.0/24	Server Network	192.168.1.1
192.168.2.0/24	Management Network	192.168.2.1

Network	Department	Gateway
192.168.3.0/24	Finance Network	192.168.3.1
192.168.4.0/24	Human Resources Network	192.168.4.1
192.168.5.0/24	IT Network	192.168.5.1

Result

pfSense successfully functioned as:

- Central routing device
- Network segmentation firewall
- Internet gateway
- Department isolation controller